Tracks, Tickets and Territories: A Spatially Heterogeneous Analysis of Train Accessibility, House Prices and Population Characteristics in Greater Melbourne

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1 Topic and Research Questions

Train stations, house prices and the people who live between them form a tightly connected spatial system. Conventional studies that ask whether rail proximity raises house prices treat the city as homogeneous and report a single, global premium. Yet the empirical literature consistently finds that this premium varies substantially in space (Dziauddin, 2019; Dziauddin et al., 2015), which is strong in some neighbourhoods, negligible (or even negative) in others. This project investigates the spatial heterogeneity for Greater Melbourne using house sales, environmental and demographic data provided in earlier assignments together with newly collected data on schools, train stations and income.

Research questions: The project addresses three inter-related questions, each examining a different facet of the same spatial system:

1. How does the relationship between proximity to a metro train station and house price vary spatially across Greater Melbourne, and which neighbourhood characteristics (income, education, family structure, school access) explain this variation?
2. Where does poor train accessibility coincide with markers of socio-economic disadvantage, indicating potential “transit deserts”?
3. Are house prices spatially clustered (and at what scale), and do these clusters align with the train network?

2 Scope

- **Geographic area.** Greater Melbourne GCCSA (ABS ASGS Edition 3), clipped using the provided Greater Melbourne (ABS) study area shapefile.
- **Spatial units.** Houses and train stations are analysed as points; suburbs (SAL) are used as polygons for aggregation and mapping; rainfall surfaces and population densities are analysed as rasters.
- **Temporal scope.** House sales; Census 2021 demographics; current metropolitan rail network (PTV GTFS, 2026). The house sales pre-date Metro Tunnel by about 8 years, so we analyse the network as it existed in 2016–2017 which keeps the time alignment between cause and effect.

3 Data Sources

The project satisfies the requirement of analysing at least one point, one polygon, and one raster dataset, as summarised in Table 1. An optional environmental covariate is listed.

Dataset Name	Type	Source
Melbourne House Sales 2016–2017	Point	provided in Assignment 2
Metro Train Stations	Point	Public Transport Victoria (2026)
Victorian Schools	Point	ACARA (ACARA, 2025)
Victorian Rainfall Stations	Point	provided in Assignment 3
ABS 2021 Census Table G02	Polygon	provided in Assignment 2
Greater Melbourne Study Area	Polygon	provided in Assignment 2
Mesh Block Population Density 100m	Raster	Australian Bureau of Statistics (2021)
Victoria Rainfall Grid (supplementary)	Raster	provided in Assignment 3

Table 1. Summary of datasets used in this project.

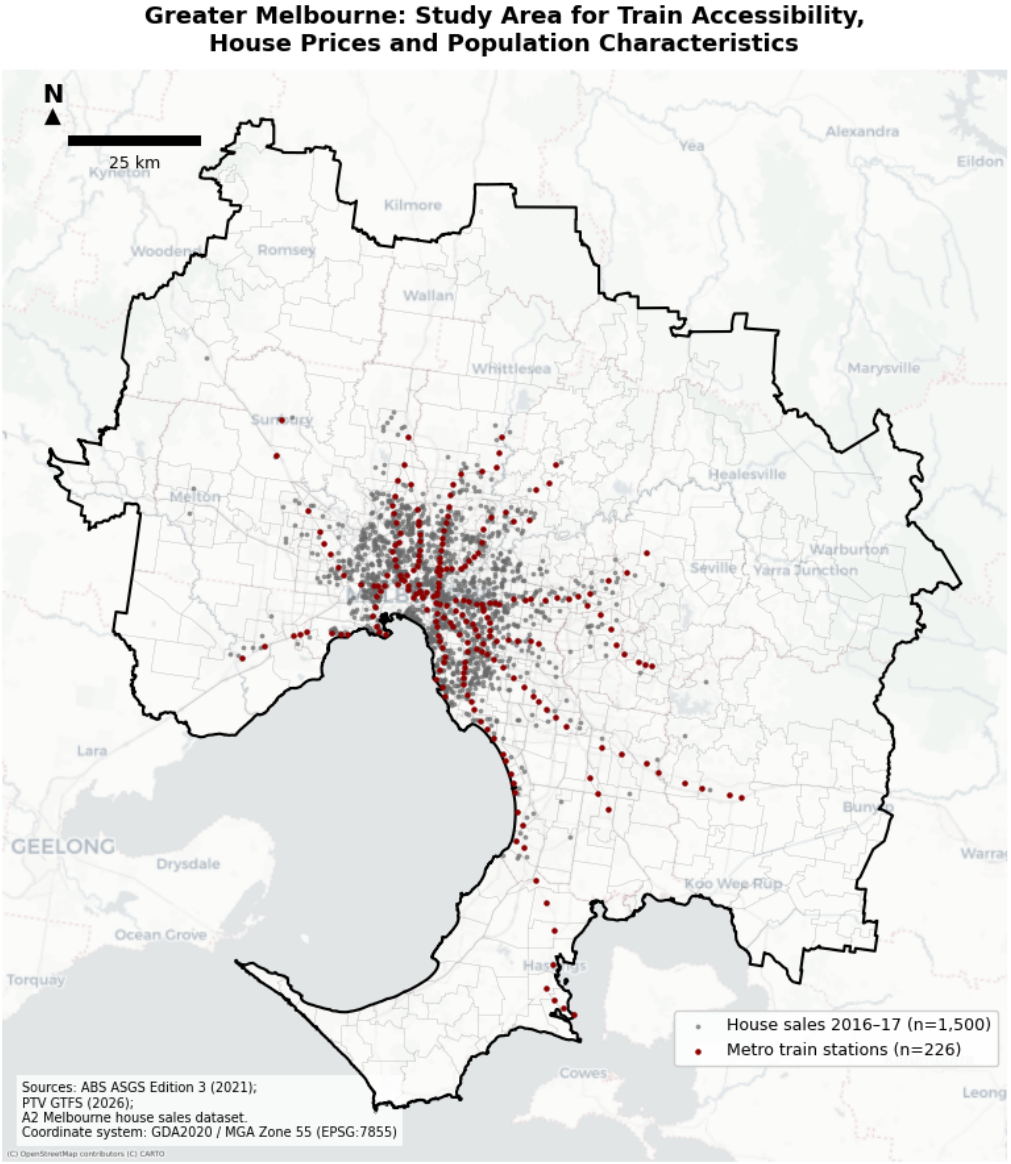


Figure 1. Study area: Greater Melbourne with the metropolitan rail network and a sample of 2016–2017 house sales overlaid on suburb boundaries. Coordinate system: GDA2020 / MGA Zone 55 (EPSG:7855).

4 Literature Review

House prices can be decomposed into the value of individual attributes, including proximity to transit (Rosen, 1974). In a recent Melbourne study, Zhang and Shukla (2023) find a substantial price premium near the Mernda heavy-rail extension. However, Liang et al. (2025) caution that effects depend on transit mode and service quality, with buses and trams in Melbourne showing very different effects from heavy rail.

A key theme in recent work is that this premium is not uniform across a city, it varies by neighbourhood and can even be negative in some areas. Dziauddin (2019) and Dziauddin et al. (2015), studying Kuala Lumpur's light rail, show that proximity is valued positively in some housing estates and negligibly or negatively in others, depending on local socio-economic context. Wang et al. (2025) report similar spatial variation in Qingdao. A parallel strand of transit equity research (Anselin, 1995; Karner, 2018) reframes the question: rather than asking what the average household gains, it asks where benefits and burdens fall.

The gap motivating our project is the absence, in Melbourne work, of a study that combines station proximity, house-price clustering and population disadvantage within a single analysis.

5 Proposed Methodology

The analysis is preceded by a standard ESDA stage (univariate summaries, pivot tables and cartographic-quality maps), extending the techniques used in previous assignments. The core analytical steps are:

1. **Environmental raster.** Mean annual rainfall is taken from the provided raster. Zonal statistics extract per-suburb means as an optional covariate.
2. **Spatial autocorrelation and equity overlay.** Global Moran's I and a Moran scatterplot establish whether sale-price values are spatially clustered at SAL level. Local Moran's I classifies suburbs into High–High, Low–Low, High–Low and Low–High quadrants; Getis–Ord G_i^* provides a complementary hot-spot perspective. A bivariate LISA of station accessibility against SEIFA disadvantage then locates clusters where the two co-vary in policy-relevant directions, operationalising the "transit desert" question.
3. **Geographically Weighted Regression.** A global OLS hedonic model establishes the baseline relationship between sale price and chosen covariates (distance to station, household income, household size, school distance, optional environmental controls). **GWR** is then fitted to the same specification to recover spatially varying coefficients, with bandwidth selected by corrected AIC. This is the project's extension beyond assignments. Residual Moran's I checks whether spatial dependence has been adequately absorbed. Multiscale GWR is fitted as a robustness check.

6 Hypothesis

We hypothesize that (i) proximity to a metro train station is associated with a statistically significant positive price premium globally; (ii) the premium varies in space, with GWR revealing larger premiums in middle-ring suburbs than in either the inner core or the far outer suburbs; and (iii) bivariate LISA will reveal outer clusters where low accessibility coincides with disadvantage.

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